Software Requirements Specification

Geospatial Disease Outbreak Mapper

Project: Geospatial Disease Outbreak Mapper
Team: Group BC
Status: Final

Document Approval

Role	Name	Signature	Date
Project Manager			
Lead Developer			
Quality Assurance			

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1. Introduction

1.1 Purpose

This document describes the requirements for the Geospatial Disease Outbreak Mapper system, which enables real-time tracking and mapping of disease outbreaks for rapid public health response.

This document is intended for:

- Development team members
- Project stakeholders
- Quality assurance testers
- System administrators
- Public health officials
- Community Health Workers (CHWs)

The system enables real-time tracking, mapping, and analysis of disease outbreaks to facilitate rapid public health response and intervention strategies.

1.2 Project Scope

1.2.1 System Overview

The Geospatial Disease Outbreak Mapper is a comprehensive mobile and web-based solution designed to:

- Enable Community Health Workers to report disease cases from field locations
- Automatically capture and validate GPS coordinates
- Provide real-time visualization of disease outbreak patterns
- Calculate and display risk assessment levels
- Support offline data collection with automatic synchronization
- Generate analytics and reports for health officials
- Maintain FHIR-compliant healthcare data standards

1.2.2 Key Benefits

- **Early Detection**: Rapid identification of disease outbreak patterns
- **Geographic Visualization**: Interactive mapping of disease hotspots
- **Data Standardization**: FHIR-compliant data for interoperability
- **Offline Capability**: Continued operation in areas with limited connectivity
- **Evidence-Based Decision Making**: Analytics for public health interventions

1.2.3 System Boundaries

In Scope:

- Mobile application for CHWs (Android/iOS)
- Web dashboard for health officials
- GPS-based location tracking
- FHIR data standardization
- Offline synchronization
- Risk assessment algorithms
- Analytics and reporting

Out of Scope:

- Direct patient treatment recommendations
- Electronic Medical Records (EMR) system integration
- Laboratory information management
- Vaccine inventory management
- Financial/billing systems

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
API	Application Programming Interface
CHW	Community Health Worker - front-line health workers who report disease cases
DFD	Data Flow Diagram
FHIR	Fast Healthcare Interoperability Resources - healthcare data exchange standard
GPS	Global Positioning System
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
REST	Representational State Transfer
SRS	Software Requirements Specification
TLS	Transport Layer Security
UI	User Interface
UML	Unified Modeling Language
WebSocket	Protocol for real-time bidirectional communication

1.4 Overview

This document is organized into eight main sections:

- **Section 2** describes the overall system context, user roles, and constraints
- **Section 3** details specific system features and functional requirements
- **Section 4** specifies external interface requirements including UI designs

- **Section 5** presents the system architecture with UML diagrams
- **Section 6** outlines non-functional requirements for performance, security, and quality
- **Section 7** covers additional requirements for database, operations, and compliance
- **Section 8** provides supplementary information and references

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2. Overall Description

2.1 Product Perspective

The Geospatial Disease Outbreak Mapper operates as a standalone mobile and web application system that integrates with:

- **GPS Services**: For automatic location capture during disease reporting
- **FHIR Server**: For healthcare data standardization and interoperability
- **Mobile Platforms**: Native support for Android and iOS devices
- **Web Browsers**: Dashboard accessible via modern web browsers
- **Cloud Infrastructure**: Scalable backend for data storage and processing

2.1.1 System Context

The system serves as a bridge between field-level disease surveillance and centralized public health monitoring, enabling:

- Data collection at the community level
- Real-time data transmission and visualization
- Evidence-based public health decision-making

2.2 Product Functions

Primary Functions:

1. **Disease Case Reporting**
    - Mobile form-based data entry
    - Automatic GPS coordinate capture
    - Patient demographic information collection
    - Disease type classification
    - Symptom documentation
2. **Real-time Outbreak Mapping**
    - Interactive geographic visualization
    - Disease case clustering
    - Multi-layer map display with risk levels

    - Filtering by disease type, date range, and location
    - Hotspot identification
3. **Data Analytics and Risk Assessment**
    - Automatic risk level calculation based on case density
    - Trend analysis and pattern detection
    - Disease distribution statistics
    - Weekly/monthly case reports
    - Predictive outbreak modeling
4. **Multi-user Role Management**
    - Role-based access control
    - CHW field reporting capabilities
    - Health official monitoring and analysis tools
    - Administrative user management
5. **Offline Synchronization**
    - Local data storage during connectivity loss
    - Automatic sync when connection restored
    - Pending case queue management
    - Conflict resolution
6. **FHIR Data Export**
    - Standardized healthcare data format
    - Interoperability with other health systems
    - Resource mapping for Observations, Encounters, and Locations

2.4 Constraints

2.4.1 Technical Constraints

- Must comply with HL7 FHIR R4 standards for data representation
- GPS functionality required for accurate location services
- Mobile data connectivity needed for real-time updates (with offline fallback)
- Minimum Android 8.0 (API level 26) or iOS 12.0 support
- Web dashboard requires modern browsers (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)

2.4.2 Regulatory Constraints

- Must comply with local health data privacy regulations
- Patient data protection and confidentiality requirements
- Healthcare data retention policies
- Cross-border data transfer restrictions (if applicable)

2.4.3 Resource Constraints

- Limited bandwidth in rural areas
- CHW smartphone hardware limitations
- Server infrastructure capacity
- Development timeline and budget

2.4.4 Operational Constraints

- System must be available 24/7 with minimal downtime
- Response time must meet user expectations
- Must support multiple concurrent users
- Data backup and recovery procedures required

2.5 Assumptions and Dependencies

2.5.1 Assumptions

1. CHWs have access to GPS-enabled smartphones
2. Health officials have reliable internet connectivity
3. GPS signals are accurate within acceptable margins (±10 meters)
4. Users have basic literacy in the system's supported languages
5. Power supply and device charging infrastructure are available
6. Mobile network coverage exists in target deployment areas (even if intermittent)
7. Healthcare organizations support FHIR data standards

2.5.2 Dependencies

1. **External Services**
   - GPS satellite availability
   - Mobile network provider infrastructure
   - Cloud hosting provider reliability
   - FHIR server availability
2. **Third-Party Components**
   - Mapping libraries (Google Maps, Mapbox, or OpenStreetMap)
   - FHIR validation libraries
   - Authentication services
   - Data visualization tools
3. **Organizational Dependencies**
   - CHW training and onboarding programs
   - Health official adoption and usage
   - IT support availability

- Ongoing maintenance funding

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3. System Features

3.1 User Authentication and Authorization

Description: Secure user login and role-based access control system

Priority: High

Stimulus/Response Sequences:

- User enters credentials → System validates → Grants role-appropriate access
- Invalid credentials → System denies access → Displays error message

Functional Requirements:

FR-001: The system shall provide a secure login interface for all users

- Input: Username/email and password
- Process: Credential validation against user database
- Output: Authentication token or error message

FR-002: The system shall implement role-based access control (RBAC)

- Roles: CHW, Public Health Official, Administrator
- Each role has specific permissions and accessible features

FR-003: The system shall maintain user session management

- Session timeout after 30 minutes of inactivity
- Secure token-based authentication
- Single sign-on support

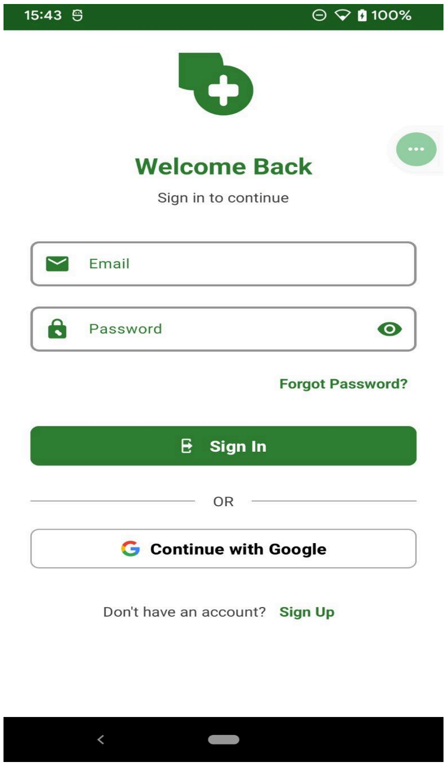
FR-004: The system shall provide password reset functionality

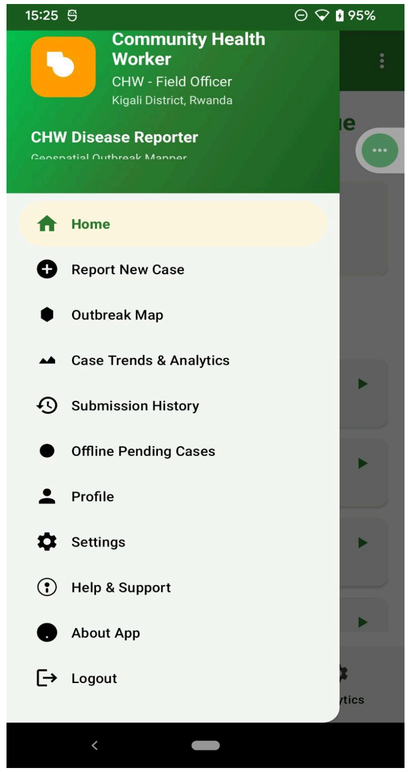
- Email-based password recovery
- Secure token generation
- Password strength validation

FR-005: The system shall log all authentication attempts

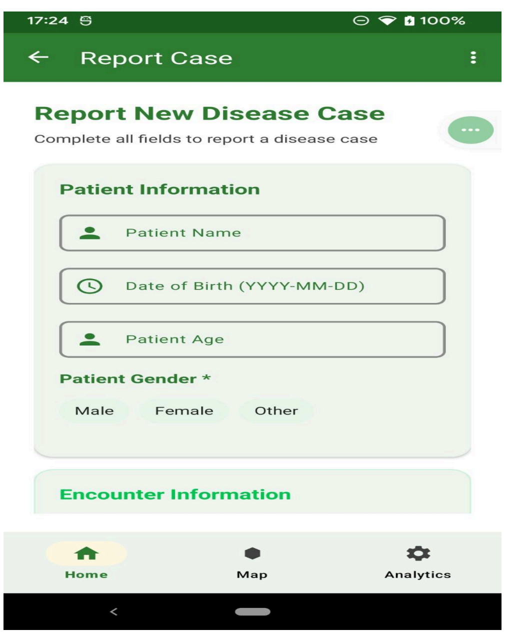
- Successful and failed login records
- Timestamp and user identification

- IP address tracking for security





3.2 Disease Case Reporting



Description: CHWs can report new disease cases through mobile application with automatic GPS capture

Priority: High

Stimulus/Response Sequences:

- CHW opens reporting form → System displays input fields
- CHW fills form → System validates data → Auto-captures GPS
- CHW submits form → System stores case → Converts to FHIR format

Functional Requirements:

FR-006: The system shall provide a disease case reporting form with the following fields:

- Patient demographic information (name, age, gender, date of birth)
- Disease type (dropdown selection)
- Symptoms and clinical observations
- Case severity
- Date of symptom onset
- Additional notes

FR-007: The system shall automatically capture GPS coordinates when a case is reported

- Latitude and longitude with accuracy metadata
- Address resolution (when connectivity available)
- Timestamp of location capture

FR-008: The system shall validate all required fields before submission

- Client-side validation for immediate feedback
- Server-side validation for data integrity
- Clear error messages for missing or invalid data

FR-009: The system shall convert disease reports to FHIR Observation resources

- Map local data structure to FHIR standard
- Generate compliant FHIR JSON
- Store both local and FHIR representations

FR-010: The system shall support offline case reporting

- Store reports locally when connectivity unavailable
- Queue for synchronization when connection restored
- Indicate sync status to user

FR-011: The system shall allow CHWs to view their submission history

- List of previously submitted cases
- Filter by date, disease type, and status
- Detailed view of individual submissions

3.3 Real-time Outbreak Mapping

Description: Health officials can view live disease outbreak maps with interactive visualization

Priority: High

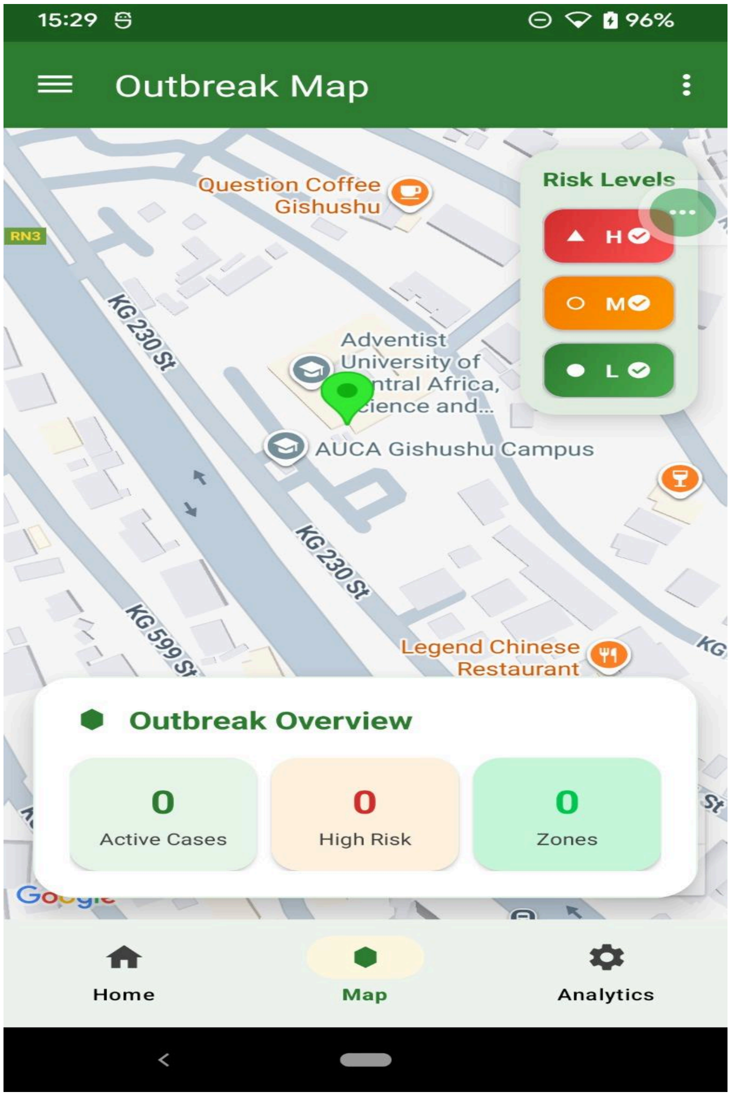
Stimulus/Response Sequences:

- User requests map view → System loads geographic data
- System displays cases on interactive map → User applies filters
- System updates visualization in real-time as new cases reported

Functional Requirements:

FR-012: The system shall display disease cases on an interactive map interface

- Marker/pin visualization for individual cases
- Color-coded by disease type or risk level
- Clickable markers for detailed case information



[INSERT IMAGE 5 HERE - Outbreak Map with Risk Levels] *Figure 3.2: Real-time Outbreak Map with Risk Level Indicators*

FR-013: The system shall implement geographic clustering for nearby cases

- Automatic grouping when multiple cases in proximity
- Cluster size indication (number of cases)
- Expandable clusters on zoom

FR-014: The system shall provide filtering capabilities

- Filter by disease type (malaria, dengue, cholera, etc.)

- Date range selection
- Risk level filtering (high, medium, low)
- Geographic boundary filtering

FR-015: The system shall display risk level indicators

- Three-tier risk classification (High, Medium, Low)
- Visual representation with color coding:
  - Red: High risk
  - Orange: Medium risk
  - Green: Low risk
- Risk level legend and explanations

FR-016: The system shall support map layer controls

- Toggle disease type layers
- Satellite/terrain/street view options
- Heatmap overlay for case density

FR-017: The system shall update map visualization in real-time

- WebSocket connection for live updates
- Notification of new cases
- Automatic refresh of risk assessments

FR-018: The system shall provide an outbreak overview panel

- Current active cases count
- High-risk zones count
- Total zones monitored
- Quick statistics summary

3.4 Risk Assessment and Analytics

Description: System automatically calculates outbreak risk levels and provides comprehensive analytics

Priority: Medium-High

Stimulus/Response Sequences:

- New case reported → System calculates risk → Updates risk assessment
- User requests analytics → System generates visualizations
- System performs periodic trend analysis → Alerts on significant patterns

Functional Requirements:

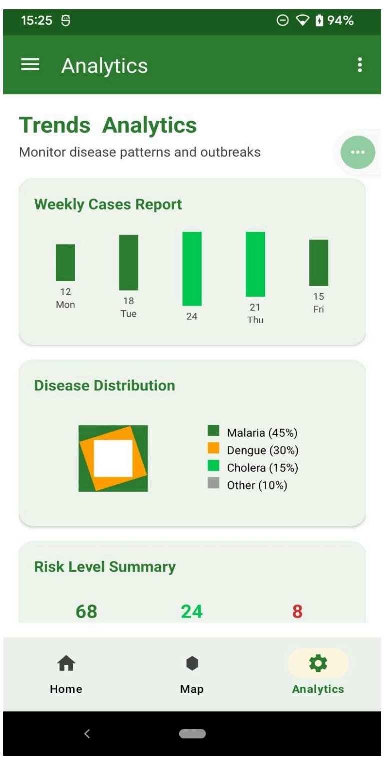
FR-019: The system shall automatically calculate risk assessment based on:

- Case density (cases per geographic area)
- Temporal clustering (cases within time window)
- Disease severity and transmission rate
- Population density in affected area

FR-020: The system shall generate FHIR RiskAssessment resources

- Link to related Observations and Encounters
- Risk probability and severity scores
- Basis for risk calculation
- Mitigation recommendations

FR-021: The system shall provide weekly case trend reports

- Bar chart visualization of cases per day
- Comparison with previous weeks
- Trend direction indicators
- 

FR-022: The system shall display disease distribution analytics

- Pie chart showing percentage breakdown by disease type
- Table view with absolute numbers
- Ability to filter by date range and location

FR-023: The system shall provide a risk level summary

- Count of low, medium, and high-risk cases
- Geographic distribution of risk levels
- Temporal trend of risk changes

FR-024: The system shall generate detailed analytical reports including:

- Case trends over time
- Geographic hotspot analysis
- Disease-specific outbreak patterns
- Demographic analysis of affected populations

FR-025: The system shall support export of analytics data

- CSV format for spreadsheet analysis
- PDF reports for documentation
- FHIR MeasureReport resources for interoperability

3.5 Offline Synchronization

Description: Mobile application supports offline data collection with automatic synchronization

Priority: High

Stimulus/Response Sequences:

- Connection lost → System switches to offline mode
- CHW reports case → Stored locally with pending status
- Connection restored → System automatically syncs pending cases
- Sync complete → User notified of successful submission

Functional Requirements:

FR-026: The system shall detect network connectivity status

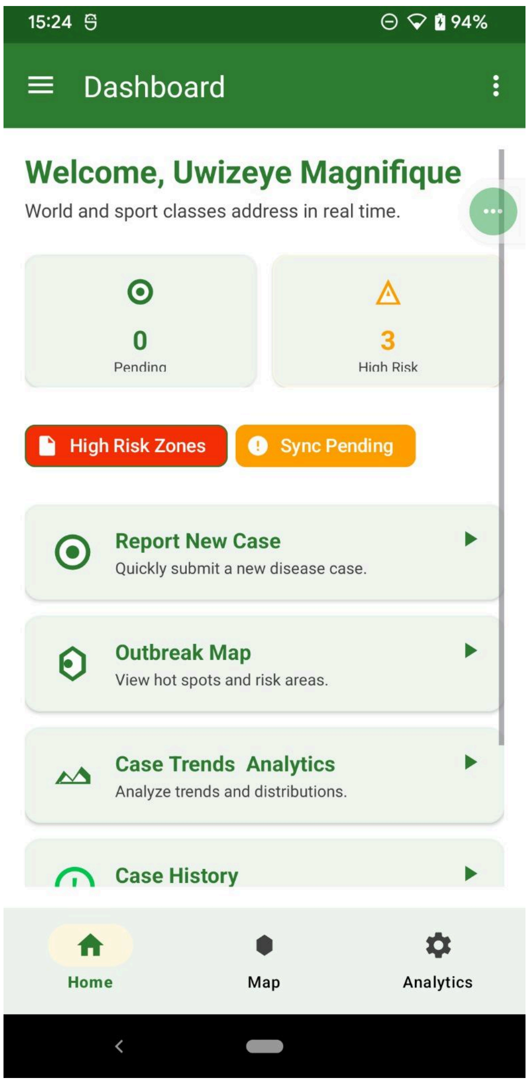
- Monitor connection state
- Indicate online/offline status to user
- Gracefully handle connection changes

FR-027: The system shall store case reports locally when offline

- SQLite database on mobile device
- Maintain data integrity and structure
- Secure local storage with encryption

FR-028: The system shall maintain a pending submission queue

- List of unsynchronized cases
- Visual indicator of pending count
- Manual sync trigger option



FR-029: The system shall automatically synchronize when connection restored

- Background sync process
- Batch submission of pending cases

- Retry logic for failed submissions

FR-030: The system shall handle sync conflicts

- Timestamp-based conflict resolution
- Preserve both versions if significant differences
- Notify administrators of conflicts requiring manual resolution

FR-031: The system shall provide sync status notifications

- Success confirmation with number of cases synced
- Error notifications with reason
- Progress indicator during sync

3.6 Report Generation

Description: Generate comprehensive reports for public health analysis and documentation

Priority: Medium

Functional Requirements:

FR-032: The system shall generate standard report types:

- Weekly epidemiological reports
- Monthly summary reports
- Disease-specific outbreak reports
- Geographic area reports

FR-033: The system shall allow custom report parameters

- Date range selection
- Disease type filtering
- Geographic area selection
- Demographic filters

FR-034: The system shall export reports in multiple formats

- PDF for distribution and archival
- Excel/CSV for data analysis
- FHIR MeasureReport for interoperability

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4. External Interface Requirements

4.1 User Interfaces

4.1.1 General UI Requirements

UIR-001: All interfaces shall follow Material Design guidelines for consistency

- Consistent color scheme: Primary green (#2E7D32), accent orange (#F57C00)
- Standard navigation patterns
- Responsive layouts for different screen sizes

UIR-002: Mobile application shall support portrait and landscape orientations

- Adaptive layouts
- Maintain functionality in both orientations

UIR-003: All text shall be clearly readable

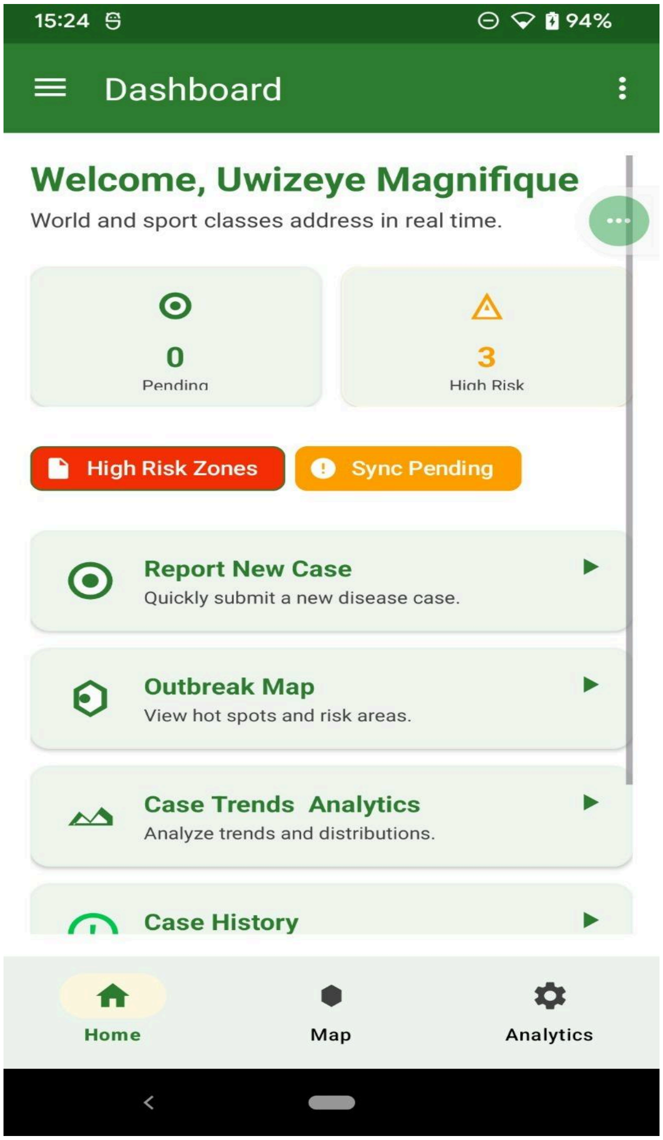
- Minimum font size: 14sp for body text
- High contrast ratios (WCAG AA compliance)
- Clear typography hierarchy

UIR-004: The system shall provide user feedback for all actions

- Loading indicators during processing
- Success/error messages
- Confirmation dialogs for critical actions

4.1.2 CHW Mobile Application Interface

Dashboard Screen



UIR-005: Dashboard shall display:

- Welcome message with CHW name and location
- Pending cases count
- High-risk cases alert
- Quick action buttons:
    - Report New Case
    - Outbreak Map
    - Case Trends Analytics
    - Case History
    - Sync Pending (when offline cases exist)

UIR-006: Navigation menu shall include:

- Home
- Report New Case
- Outbreak Map
- Case Trends & Analytics
- Submission History
- Offline Pending Cases
- Profile
- Settings
- Help & Support
- About App
- Logout

Disease Reporting Form

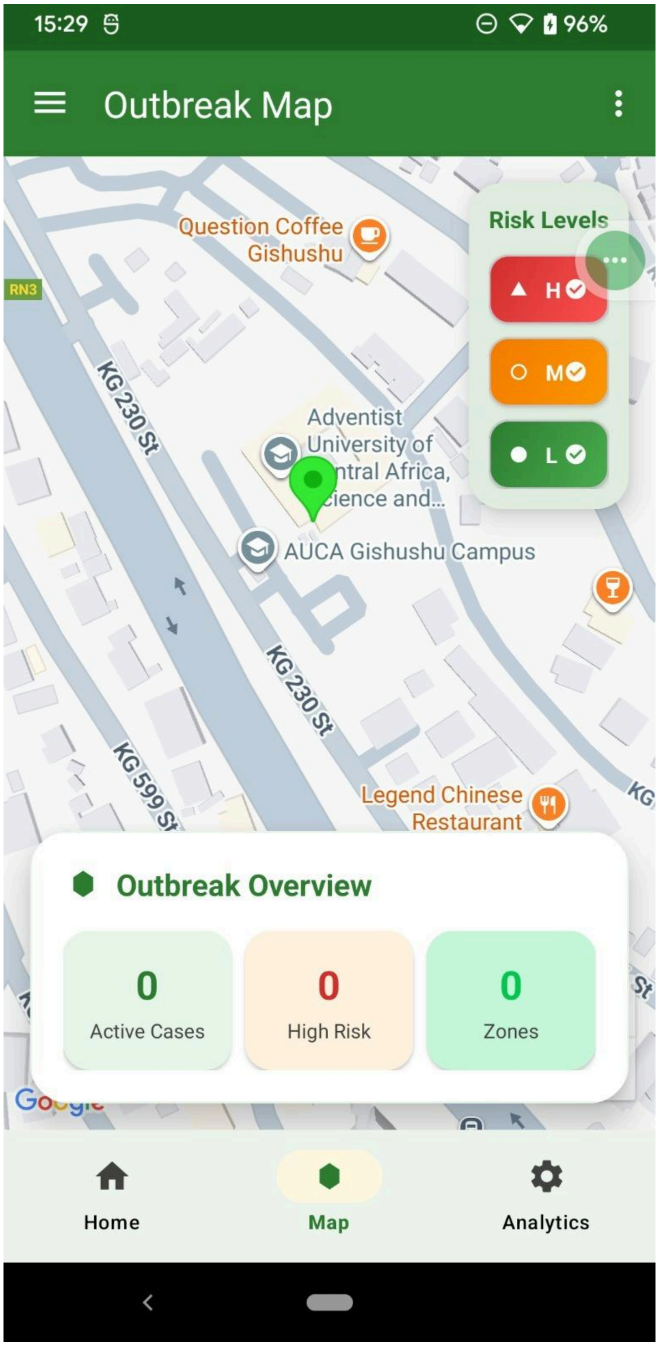
UIR-007: Reporting form shall include grouped sections:

- Patient Information
  - Patient ID (auto-generated or manual)
  - Name (optional for privacy)
  - Age/Date of Birth
  - Gender selection
- Clinical Information
  - Disease type dropdown
  - Symptoms checklist
  - Severity level selector
  - Date of symptom onset
- Location Information
  - Auto-captured GPS coordinates (display lat/long)
  - Manual address entry option
  - Location accuracy indicator
- Additional Information
  - Notes text area
  - Photo attachment option (if needed)

UIR-008: Form validation requirements:

- Required field indicators (*)
- Real-time validation with inline error messages
- Clear submit button (enabled only when valid)
- Draft save functionality

Outbreak Map View



UIR-009: Map interface shall include:

- Full-screen interactive map
- Risk level legend (High/Medium/Low with color indicators)
- Disease markers with clustering

- Bottom sheet with outbreak overview:
    - Active Cases count
    - High Risk count
    - Zones count
- Filter button for disease types and date ranges
- User location indicator

UIR-010: Analytics screen shall display:

- Weekly Cases Report (bar chart)
- Disease Distribution (pie chart with legend)
- Risk Level Summary (color-coded statistics)
- Scrollable content for additional metrics
- Date range selector
- Export options

4.1.3 Health Official Web Dashboard

UIR-011: Web dashboard shall feature:

- Responsive layout for desktop and tablet
- Sidebar navigation
- Main content area with widgets
- Real-time data updates

UIR-012: Dashboard widgets shall include:

- Live outbreak map (large central widget)
- Statistics cards:
    - Total active cases
    - New cases today
    - High-risk zones
    - Trending diseases
- Recent activity feed
- Quick filters and search

UIR-013: Map visualization shall provide:

- Full-featured interactive map
- Multiple layer controls

- Advanced filtering options
- Export map as image functionality
- Print-friendly view

UIR-014: Analytics section shall offer:

- Comprehensive charts and graphs
- Customizable dashboards
- Drill-down capabilities
- Export to various formats
- Scheduled report generation

4.2 Hardware Interfaces

4.2.1 GPS Receiver Interface

HIR-001: The system shall interface with device GPS hardware

- Access to GPS/GNSS receivers
- Support for A-GPS (Assisted GPS) for faster lock
- Fallback to network-based location (WiFi/Cell tower triangulation)

HIR-002: GPS accuracy requirements:

- Target accuracy: ≤10 meters
- Minimum acceptable accuracy: ≤50 meters
- Display accuracy metadata to users
- Alert if accuracy exceeds acceptable threshold

HIR-003: The system shall handle GPS unavailability

- Allow manual location entry
- Save cases with "GPS unavailable" flag
- Attempt GPS capture in background

4.2.2 Mobile Device Interface

HIR-004: Touch screen interaction requirements:

- Support for standard touch gestures (tap, swipe, pinch-to-zoom)
- Minimum touch target size: 48x48 dp
- Haptic feedback for important actions

HIR-005: Camera interface (if photo capture enabled):

- Access to device camera
- Image compression before upload
- Secure storage of images

HIR-006: Storage interface:

- Access to internal storage for offline data
- Minimum 100MB free space required
- Storage space monitoring and cleanup

4.2.3 Network Connectivity

HIR-007: The system shall support multiple connectivity types:

- WiFi (802.11 b/g/n/ac)
- Mobile data (3G/4G/LTE/5G)
- Automatic switching between connection types
- Bandwidth-appropriate data transfer

4.3 Software Interfaces

4.3.1 Operating System Interfaces

SIR-001: Android Platform Requirements:

- Minimum: Android 8.0 (API level 26)
- Target: Latest stable Android version
- Support for Android system services:
    - Location Services
    - Network Manager
    - Notification Manager
    - Background Services

SIR-002: iOS Platform Requirements:

- Minimum: iOS 12.0
- Target: Latest stable iOS version
- Integration with iOS frameworks:
    - Core Location
    - MapKit

    - Core Data

4.3.2 Backend Services Interface

SIR-003: FHIR Server Interface:

- Protocol: HTTPS/REST
- FHIR Version: R4
- Supported resources:
    - Observation (disease cases)
    - Encounter (CHW-patient interactions)
    - Location (GPS data)
    - Patient (demographic data)
    - RiskAssessment (outbreak risk)
    - MeasureReport (analytics)

SIR-004: Resource interactions:

- CREATE: POST new resources
- READ: GET individual resources
- UPDATE: PUT resource modifications
- SEARCH: GET with query parameters
- Batch operations: POST Bundle resources

4.3.3 Third-Party API Interfaces

SIR-005: Mapping Service API:

- Provider: Google Maps, Mapbox, or OpenStreetMap
- Functions:
    - Map tile rendering
    - Geocoding (address to coordinates)
    - Reverse geocoding (coordinates to address)
    - Route calculation (if needed)

SIR-006: Authentication Service:

- JWT (JSON Web Token) based authentication
- OAuth 2.0 support for future integrations
- Refresh token mechanism

4.4 Communications Interfaces

4.4.1 Network Protocols

CIR-001: HTTP/HTTPS Protocol:

- All API communication via HTTPS
- TLS 1.2 or higher
- Certificate validation required
- HTTP/2 support for improved performance

CIR-002: WebSocket Protocol:

- Real-time updates for map and dashboard
- WSS (WebSocket Secure) for encrypted communication
- Automatic reconnection on connection loss
- Heartbeat mechanism to maintain connection

4.4.2 Data Formats

CIR-003: JSON Data Format:

- Primary data exchange format
- UTF-8 encoding
- Schema validation
- Compression (gzip) for large payloads

CIR-004: FHIR JSON Format:

- Strict adherence to FHIR R4 specification
- Resource validation before transmission
- Support for contained resources
- Bundle support for batch operations

4.4.3 API Endpoints

CIR-005: RESTful API Structure:

Base URL: https://api.disease-mapper.example.com/v1

Authentication:
- POST /auth/login
- POST /auth/logout

- POST /auth/refresh

Disease Reports:
- POST /reports
- GET /reports
- GET /reports/{id}
- PUT /reports/{id}

Analytics:
- GET /analytics/trends
- GET /analytics/distribution
- GET /analytics/risk-summary

Map Data:
- GET /map/cases
- GET /map/clusters
- GET /map/risk-zones

User Management:
- GET /users/profile
- PUT /users/profile
- POST /users/change-password

Sync:
- POST /sync/cases (batch upload)
- GET /sync/status

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5. System Architecture

5.1.1 Actor Descriptions

Community Health Worker (CHW):

- Primary actor for field-level data collection
- Reports disease cases from community locations
- Conducts screening and patient interviews
- Accesses local health information and resources
- Provides follow-up care and sends referrals
- Views submission history and case status
- Operates primarily through mobile application

Public Health Official:

- Secondary actor for monitoring and analysis
- Views outbreak maps and analytics
- Generates reports for decision-making
- Assesses risk levels across regions
- Identifies intervention priorities
- Accesses comprehensive dashboards
- Operates primarily through web interface

System (Automated Actor):

- Auto-captures GPS coordinates
- Performs data validation
- Converts data to FHIR format
- Calculates risk assessments
- Generates notifications and alerts
- Manages data synchronization
- Executes scheduled tasks and backups

5.1.2 Use Case Descriptions

UC-001: Conduct Screening

- Actor: CHW
- Description: CHW conducts health screening and collects patient information
- Precondition: CHW is authenticated and has mobile device
- Main Flow:
    1. CHW initiates screening process
    2. System displays screening form
    3. CHW enters patient information
    4. System validates input
    5. CHW submits screening data
    6. System captures GPS and stores data
- Postcondition: Screening record created and stored

UC-002: Report Disease Case (see Section 3.2 for detailed requirements)

UC-003: View Outbreak Map (see Section 3.3 for detailed requirements)

UC-004: Access Resilience Toolkit

- Actor: CHW
- Description: Access health resources and guidelines
- Precondition: CHW authenticated
- Main Flow:
    1. CHW selects toolkit option
    2. System displays available resources

    3. CHW browses and downloads materials
- Postcondition: Resources accessed

UC-005: Sync Offline Data

- Actor: CHW, System
- Description: Synchronize locally stored data with server
- Precondition: Device was offline, now has connectivity
- Main Flow:
    1. System detects network connection
    2. System initiates sync process
    3. Pending cases uploaded in batch
    4. Server validates and stores data
    5. System updates local status
    6. User notified of sync completion
- Postcondition: All pending data synchronized

UC-006: Send Referrals

- Actor: CHW
- Description: Refer patients to healthcare facilities
- Precondition: Patient requires additional care
- Main Flow:
    1. CHW creates referral request
    2. System captures referral details
    3. Referral sent to designated facility
    4. System tracks referral status
- Postcondition: Referral recorded and transmitted

UC-007: Provide Follow-up Care

- Actor: CHW
- Description: Document follow-up visits and patient progress
- Precondition: Previous case exists
- Main Flow:
    1. CHW selects patient from case history
    2. System displays patient record
    3. CHW documents follow-up information
    4. System updates case status
- Postcondition: Follow-up documented

UC-008: Exchange Reports & Analytics

- Actor: CHW, Public Health Official, FHIR Server
- Description: Generate and share analytical reports
- Precondition: Data available for analysis

- Main Flow:
    1. User requests report generation
    2. System queries database
    3. Analytics calculated and visualized
    4. Report exported in desired format
    5. FHIR MeasureReport resources generated
    6. Reports shared with FHIR server
- Postcondition: Reports generated and available

5.2 Class Diagram

5.2.1 Key Class Descriptions

CHW (Community Health Worker)

- Attributes:
    - ID: String
    - Name: String
    - Role: String
- Relationships:
    - Creates Encounters (1 to many)
    - Submits DiseaseReports (1 to many)
    - Generates MeasureReports (1 to many)

Patient

- Attributes:
    - PatientID: String
    - Name: String
    - DateOfBirth: Date
    - Gender: String
- Relationships:
    - Participates in Encounters (1 to many)

Encounter

- Attributes:
    - EncounterID: String
    - EncounterDate: DateTime
    - EncounterType: String
- Relationships:
    - Involves one CHW (many to 1)
    - Involves one Patient (many to 1)

    - Associated with GPSLocation (0 to 1)
    - Generates DiseaseReport (1 to 1)
    - Produces Observation (1 to many)

DiseaseReport

- Attributes:
    - ReportID: String
    - DiseaseType: String
    - ReportDate: Date
    - Status: String
- Relationships:
    - Generated from Encounter (1 to 1)
    - Has GPSLocation (1 to 1)
    - Assessed by RiskAssessment (1 to 1)

GPSLocation

- Attributes:
    - Latitude: Decimal
    - Longitude: Decimal
    - Address: String
- Relationships:
    - Associated with Encounters
    - Linked to DiseaseReports

Observation

- Attributes:
    - ObservationID: String
    - Details: String
    - TimeStamp: DateTime
- Relationships:
    - Recorded during Encounter (many to 1)

RiskAssessment

- Attributes:
    - RiskID: String
    - Level: String (High/Medium/Low)
    - Description: String
- Relationships:
    - Evaluates DiseaseReport (1 to 1)

MeasureReport

- Attributes:
  - MeasureID: String
  - ReportType: String
  - PeriodStart: Date
  - PeriodEnd: Date
  - CaseCount: Integer
- Relationships:
  - Created by CHW or System
  - Aggregates multiple DiseaseReports

5.3 Data Flow Diagrams

5.3.1 Level 1 Data Flow Diagram

Processes:

1. Conduct Screening

- Input: Patient information from CHW
- Output: Screening data to "Manage Patient Records"
- Description: Initial patient health assessment

2. Manage Patient Records

- Input: Screening data, encounter information
- Output: Updated records to data stores
- Interaction: D1 (Patient Records), D2 (Encounter Records)
- Description: Central patient data management

3. Send Referrals

- Input: Referral requests from screening
- Output: Referrals to higher-level care facilities
- Description: Patient referral management

4. Provide Follow-up Care

- Input: Previous encounter data
- Output: Updated follow-up records
- Interaction: D2 (Encounter Records)
- Description: Ongoing patient care documentation

5. Exchange Reports & Analytics

- Input: Aggregated data from all data stores
- Output: Reports to FHIR Server, D4 (Reports & Analytics)
- Interaction: D1, D2, D3, D4
- Description: Generate analytical reports and share via FHIR

5.4 Sequence Diagrams

Interaction Flow:

1. User enters login credentials in the User Interface
2. System validates user credentials
3. System verifies credentials with Database
4. Database returns user records to System
5. System requests system data from Database
6. Database fetches and returns records
7. System sends "Login Successful" message to User
8. System displays reports to User

Actors:

- User (CHW, Health Official, or Admin)
- System (Application Logic)
- Database (Data Storage)
- Admin (for administrative oversight)

5.4.2 Disease Reporting Sequence

Interaction Flow:

1. CHW opens disease reporting form
2. System displays form interface
3. CHW enters patient and disease information
4. System validates input fields
5. System captures GPS coordinates
6. CHW submits form
7. System creates DiseaseReport object
8. System generates FHIR Observation resource
9. System stores data in database
10. System calculates risk assessment
11. System returns confirmation to CHW
12. System notifies health officials of new case

5.4.3 Map Update Sequence

Interaction Flow:

1. New disease case reported and stored
2. System triggers map update event
3. WebSocket connection pushes update to connected clients
4. Client receives new case data
5. Client updates map visualization
6. Client displays new marker on map
7. Client updates statistics and risk indicators

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6. Non-Functional Requirements

6.1 Performance Requirements

PER-001: Response Time

- Map should load within 3 seconds on standard network connection
- Case report submission should complete within 5 seconds
- Dashboard should refresh within 2 seconds
- Search queries should return results within 1 second
- Authentication should complete within 2 seconds

PER-002: Throughput

- System should support 1000+ concurrent users
- Handle 500 case submissions per hour during peak times
- Process 100 map requests per minute
- Support 50 simultaneous report generations

PER-003: Resource Usage

- Mobile app size: < 50MB
- Mobile app memory usage: < 200MB RAM
- Database query response: < 500ms for 95% of queries
- API response time: < 300ms (excluding network latency)

PER-004: Scalability

- Horizontal scaling capability for backend services

- Support for 10,000+ disease cases in database
- Handle 100,000+ map markers efficiently through clustering
- Accommodate growth to 5,000+ registered users

PER-005: Data Synchronization

- Offline data sync within 30 seconds of connection restoration
- Batch sync of up to 100 cases simultaneously
- Background sync without impacting foreground operations

6.2 Safety Requirements

SAF-001: System Reliability

- No single point of failure for emergency disease reporting
- Redundant server infrastructure with automatic failover
- Regular health checks and monitoring
- Automatic system recovery procedures

SAF-002: Data Backup and Recovery

- Daily automated database backups
- Point-in-time recovery capability (up to 30 days)
- Backup storage in geographically separate location
- Recovery Time Objective (RTO): < 4 hours
- Recovery Point Objective (RPO): < 1 hour

SAF-003: Offline Data Protection

- Local data encrypted on mobile device
- Automatic local backup of offline data
- Data integrity checks before synchronization
- Protection against data loss during device failure

SAF-004: Error Handling

- Graceful degradation of services
- User-friendly error messages
- Detailed error logging for troubleshooting
- Automatic retry mechanisms for transient failures

6.3 Security Requirements

SEC-001: Data Encryption

- All data encrypted in transit using HTTPS/TLS 1.2+
- Patient data encrypted at rest (AES-256)
- Secure key management and rotation
- End-to-end encryption for sensitive communications

SEC-002: Authentication and Authorization

- Multi-factor authentication option for administrators
- Password complexity requirements (min 8 characters, mixed case, numbers, special characters)
- Account lockout after 5 failed login attempts
- Role-based access control (RBAC) enforcement
- Session timeout after 30 minutes of inactivity

SEC-003: Data Privacy

- Compliance with healthcare data privacy regulations
- Patient data anonymization options
- Access logging for all patient data views
- Data minimization principles
- Right to erasure implementation (GDPR compliance if applicable)

SEC-004: Audit and Logging

- Comprehensive audit trails for all data modifications
- Log retention for minimum 1 year
- Tamper-proof logging mechanism
- Security event monitoring and alerting
- Regular security audit reviews

SEC-005: Input Validation

- Server-side validation for all user inputs
- SQL injection prevention
- Cross-Site Scripting (XSS) protection
- CSRF token implementation
- File upload restrictions and scanning

SEC-006: API Security

- Rate limiting to prevent abuse (100 requests/minute per user)
- API authentication via JWT tokens
- Token expiration and refresh mechanisms
- IP-based access restrictions for admin endpoints

6.4 Software Quality Attributes

6.4.1 Availability

QA-001: System uptime requirement: 99.5% (approximately 43 hours downtime per year)

- Scheduled maintenance windows during off-peak hours
- Notification of planned downtime at least 48 hours in advance
- Maximum unplanned downtime per incident: 2 hours

6.4.2 Reliability

QA-002: Offline capability for mobile application

- Full functionality for data collection when offline
- Reliable data persistence in local storage
- Conflict-free synchronization mechanism
- Data consistency guarantee across devices

QA-003: Mean Time Between Failures (MTBF): > 720 hours **QA-004:** Mean Time To Repair (MTTR): < 2 hours

6.4.3 Maintainability

QA-005: Modular architecture with clear separation of concerns

- Well-documented codebase with inline comments
- API documentation (OpenAPI/Swagger specification)
- Database schema documentation
- Deployment and configuration guides

QA-006: Code quality standards

- Unit test coverage: > 80%
- Integration test coverage for critical paths
- Automated code quality checks (linting, static analysis)
- Code review requirements for all changes

QA-007: Monitoring and diagnostics

- Application performance monitoring (APM)
- Error tracking and reporting
- Health check endpoints
- System resource monitoring

6.4.4 Portability

QA-008: Platform support

- Android 8.0+ support (covers 95%+ of Android devices)
- iOS 12.0+ support (covers 95%+ of iOS devices)
- Web browser compatibility:
  - Chrome 90+
  - Firefox 88+
  - Safari 14+
  - Edge 90+

QA-009: Data portability

- FHIR-compliant export functionality
- Standard CSV/Excel export for analytics
- Database migration tools
- Import functionality from other systems

6.4.5 Usability

QA-010: User interface design

- Intuitive navigation requiring minimal training
- Consistent design patterns across platforms
- Accessibility compliance (WCAG 2.1 Level AA)
- Support for multiple languages (localization ready)

QA-011: User assistance

- Context-sensitive help documentation
- In-app tutorials for first-time users
- Error messages with actionable guidance
- User manual and video tutorials

QA-012: Task efficiency

- Average time to report a case: < 3 minutes
- Dashboard information at-a-glance design
- Quick filters and search functionality
- Keyboard shortcuts for web interface

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7. Other Requirements

7.1 Database Requirements

DB-001: Data Storage

- FHIR-compliant data schema
- Support for structured (patient demographics) and semi-structured data (clinical observations)
- Geospatial data types for GPS coordinates
- Temporal data for case tracking and trends

DB-002: Spatial Indexing

- Geographic indexing for efficient location-based queries
- Support for proximity searches (cases within radius)
- Polygon/boundary-based queries for region analysis
- Optimized clustering algorithms for map display

DB-003: Performance Optimization

- Indexed columns for frequently queried fields (disease type, date, location)
- Query optimization for analytical reports
- Database connection pooling
- Caching layer for frequently accessed data

DB-004: Data Integrity

- Foreign key constraints to maintain referential integrity
- Data validation constraints (check constraints)
- Transaction support for multi-step operations
- Atomic operations for critical data modifications

DB-005: Audit Trail

- Comprehensive change tracking for all records
- Timestamp and user identification for all modifications
- Soft delete capability (mark as deleted vs. permanent deletion)
- Version history for critical records

7.2 Operations Requirements

OP-001: Backup Procedures

- Automated daily full database backups
- Hourly incremental backups during business hours

- Transaction log backups every 15 minutes
- Backup verification and restoration testing quarterly
- Off-site backup storage with encryption

OP-002: Monitoring and Alerting

- 24/7 system monitoring
- Real-time alerts for critical issues:
    - System downtime
    - Database connection failures
    - High error rates (> 5% of requests)
    - Disk space < 20% free
    - CPU/Memory usage > 90%
- Performance metric tracking (response times, throughput)
- User activity monitoring

OP-003: Maintenance Procedures

- Scheduled maintenance windows: Sundays 02:00-04:00 (local time)
- Database optimization (index rebuilding, statistics update) weekly
- Log rotation and archival monthly
- Security patch deployment within 48 hours of release
- Feature updates deployed quarterly (or as needed)

OP-004: Support and Helpdesk

- Tiered support system:
    - Tier 1: User assistance (CHWs, health officials)
    - Tier 2: Technical support (system issues)
    - Tier 3: Developer support (bug fixes, critical issues)
- Support availability: 8:00-18:00 local time on weekdays
- Emergency hotline for critical issues (24/7)
- Ticketing system for issue tracking
- Response time SLAs:
    - Critical: 1 hour
    - High: 4 hours
    - Medium: 1 business day
    - Low: 3 business days

OP-005: Regular Security Updates

- Monthly security reviews
- Quarterly penetration testing
- Annual comprehensive security audit
- Vulnerability scanning (weekly automated scans)
- Immediate patching of critical vulnerabilities

OP-006: Disaster Recovery

- Documented disaster recovery plan
- Annual disaster recovery drills
- Backup restoration procedures
- Failover testing semi-annually
- Communication plan for stakeholders

7.3 Legal and Regulatory Requirements

LEG-001: Health Data Privacy Compliance

- Compliance with local health data protection regulations
- Patient consent management for data collection
- Data anonymization capabilities for research purposes
- Right to access, rectify, and delete personal data
- Data breach notification procedures (within 72 hours)

LEG-002: Data Retention

- Minimum retention period: 7 years for health records
- Automated data archival after retention period
- Secure data deletion procedures
- Retention policy documentation and enforcement

LEG-003: Interoperability Standards

- HL7 FHIR R4 compliance
- Support for standard healthcare terminologies (SNOMED CT, LOINC, ICD-10)
- ISO 13606 compliance for electronic health records (if applicable)
- OpenAPI specification for API documentation

LEG-004: Accessibility Requirements

- WCAG 2.1 Level AA compliance
- Support for screen readers
- Keyboard navigation support
- Color contrast requirements
- Alternative text for images
- Resizable text without loss of functionality

LEG-005: User Agreement and Terms of Service

- Clear terms of service document

- Privacy policy readily accessible
- User consent for data collection and processing
- Cookie policy (for web interface)
- Regular review and update of legal documents

LEG-006: Geographic and Jurisdictional Compliance

- Compliance with Rwanda health regulations (primary deployment)
- Cross-border data transfer agreements (if applicable)
- Local language support (English, Kinyarwanda, French)
- Cultural sensitivity in user interface and communications

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5. ISO/IEC 25010:2011 - Systems and software Quality Requirements and Evaluation